

Cassandra Query Language (CQL). For Java (JDBC) and Python (DBAPI2), language drivers are available. Cassandra is a hybrid between a key-value and a tabular database. Each key in Cassandra is mapped to a value, which is an object; values as columns, and columns are grouped into sets, which are known as column families. Column families have rows and columns, where each row has a row key and multiple columns. Column families can be grouped in super column families. Cassandra supports client libraries for Python, Java, .NET, Ruby, PHP, Perl and C++. It is integrated with Ganglia and has a plug-in for Nagios as well, for monitoring purpose.

Management

7. Deltacloud

Deltacloud is an API written in Ruby whose stable version is 1.0.3, developed by Red Hat and the Apache Software Foundation, which abstracts differences between various cloud offerings. It was first announced on September 3, 2009.

Each IaaS provider such as AWS, Rackspace, Eucalyptus, VMware, OpenStack, CloudStack etc, provides its own cloud API. Different APIs are difficult to deal with to manage diverse cloud offerings, and thus Deltacloud provides relief with a unified REST-based API, which can be used to manage services on any cloud. It provides an API server and the drivers necessary to connect to cloud service providers. Amazon EC2, IBM SmartCloud, GoGrid, OpenNebula, Rackspace, Eucalyptus, VMware, OpenStack and others are supported providers.

8. JCLOUDS

JCLOUDS is a cloud interface, an open source library, which can be used by experts with Java and Clojure development skills. It offers BlobStore and ComputeService API abstractions as Java libraries. It uses concepts such as maps to have a familiar programming model. It provides location-aware abstractions considering ISO-3166 codes to identify which country a cloud runs in. It provides asynchronous commands and encryption modules to customise configuration to match performance and security needs.

JCLOUDS enjoys the support of more than 25 cloud providers. AWS-Cloudwatch, AWS-EC2, Cloudstack, Deltacloud, Eucalyptus (generic), GoGrid, OpenStack-Nova, OpenStack-Nova-EC2, and VCloud (generic) are the providers supported by the API.

9. Libcloud

Libcloud is a Python library for multi-cloud management, which abstracts away differences among multiple cloud provider APIs. It was initiated by CloudKick, and now is a part of Apache Incubator. Libcloud is composed of multiple self-sustainable components, such as Compute-Cloud Servers, services such as Rackspace and AWS, storage services such as Rackspace Cloud Files and AWS S3, Load Balancers as a Service, and DNS as a Service. Each component provides a

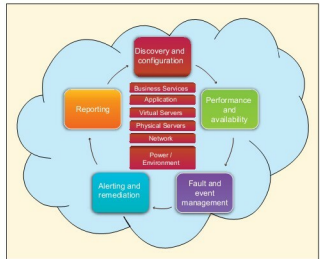


Figure 2: Zenoss

simple and easy-to-use API, which works well with supported cloud providers. AWS, OpenStack, OpenNebula, GoGrid, IBM, VMware, etc, are some of the supported providers. Libcloud supports Python 2.5, Python 2.6, Python 2.7, PyPy, and Python 3 (since 0.7.1).

Monitoring

10. Zenoss

Zenoss is an open source application, server, and network management platform released under the GNU General Public License (GPL) version 2, based on the Zope application server. Zenoss Core consists of open source technologies such as the object-oriented Web server, Zope Application server; an extensible programming language (Python); the monitoring protocol Net-SNMP that collects system status details; graph and log time-series data by RRDtool; open source database, MySQL; and the event-driven networking engine, Twisted. It provides an easy-to-use Web interface to monitor performance, events, configuration and inventory. There are a number of products and tools in the market, which provide monitoring of on-premise/cloud environments. Zenoss is one of the best for unified monitoring, since it is cloud agnostic.

Zenoss monitors networks, servers, HVAC and power, and applications using agent-less technology. Zenoss involves configuration, performance, availability, fault and event management, alerts and remediation, as well as reporting. It supports integration using Web Service APIs (XML-RPC and REST), XML import/export and a Python Scripting Shell. 

By: Mitesh Soni

The author is a technical lead at iGATE. He is in the Cloud Services (Research & Innovation) Group and loves to write about new technologies. Blog: <http://clean-clouds.com>